

Portfolio Optimization via Multi-Asset Diversification: A Quantitative Backtest (2021-2026)

Author: Anish Vinod

Technical Collaborator: Gemini (AI Co-Pilot)

Date: May 3, 2026

1. Executive Summary

This research evaluates the effectiveness of a multi-asset investment strategy compared to the Nifty 50. Using Python and assisted by **Gemini (Large Language Model)** for technical implementation and debugging, this study demonstrates that a balanced allocation between Indian Equities, US Technology, and Gold significantly outperforms a "Home Bias" equity strategy.

2. Methodology & Asset Allocation

The backtest was conducted using the **yfinance** API. Gemini acted as the technical co-pilot, assisting in script architecture, handling library updates (e.g., transitioning to Adjusted Close parameters), and calculating risk metrics.

- Benchmark:** Nifty 50 Index (^NSEI) — 100% Allocation.
- Strategy:** 40% Nifty 50, 30% Nasdaq-100 (QQQ), 30% Gold (GC=F).

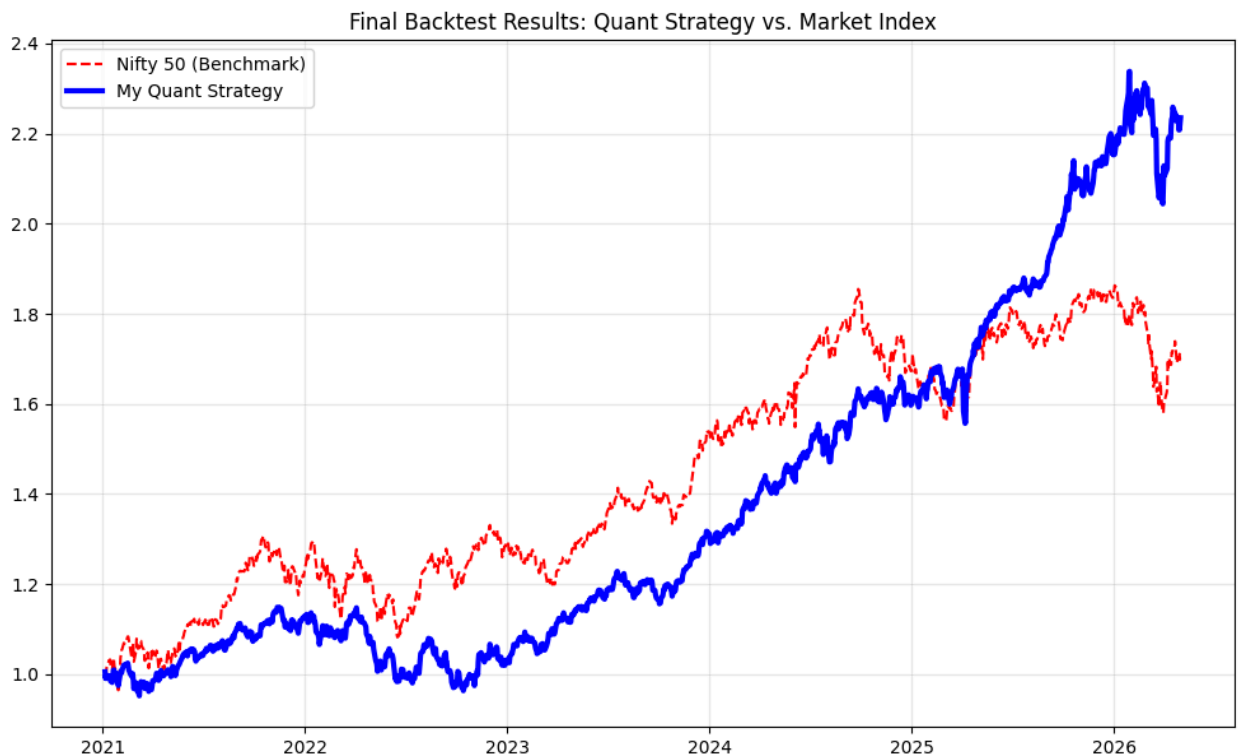
3. Quantitative Results

Below is the performance comparison generated by the backtesting engine:

Metric	My Portfolio	Nifty 50
Total Return	124.0%	70.0%
Annualized Return	15.35%	10.64%
Annualized Risk	11.56%	13.95%
Sharpe Ratio	0.90	0.40

4. Visual Analysis

The chart below illustrates the growth of 1 unit of capital from Jan 2021 to May 2026.



5. Key Observations

- **Alpha Generation:** The strategy generated a 124% total return, resulting in a 54% outperformance over the benchmark.
- **Risk-Adjusted Efficiency:** With a **Sharpe Ratio of 0.90** (vs. 0.40), the strategy proved twice as efficient as the index.
- **The "AI-Co-Pilot" Factor:** By utilizing Gemini for the heavy lifting of Python coding, the researcher was able to focus entirely on **Asset Selection** and **Strategic Interpretation**, proving that AI-human collaboration is the future of Quantitative Research.

6. Conclusion

A multi-asset strategy is mathematically superior to single-index investing. This project confirms that a 16-year-old with strategic intent and AI-augmented tools can produce professional-grade financial analysis.